Jim Eckerlein
Software Engineer

Berlin, Germany
Email: jim.eckerlein@gmail.com
Web: jimec.dev
GitHub: github.com/jim-ec

EXPERIENCE

GPU Software Engineer, UX3D GmbH
January 2019 — Present

- Concept, architecture, and development of complex software solutions for web, desktop, and embedded systems
- Development and optimization of modern 3D rendering technologies, graphics engines, and shaders
- Design and implementation of full-stack applications as well as scalable backend architectures
- Development and integration of innovative software solutions using AI-assisted development methods
- Implementation and continuous improvement of open 3D standards and associated software tools
- Independent technical project lead and management of international clients
- Technical consulting, client and account management, including contract preparation and invoicing
- Technical mentoring and support for team members on complex technical challenges
- Software development with C++, C, Qt, Rust, JavaScript, TypeScript, React, Svelte, and Python, as well as modern graphics APIs such as WebGL, Vulkan, DirectX, Metal, WebGPU, and CUDA

Reference

Software Developer, MBS Electronic Systems GmbH & Co. KG
June 2017 — December 2018

- Implementation of a PDF rendering widget in C++, QML, Qt Quick targeting an embedded device

Reference

Trainee, ESR Labs GmbH
September 2015 — July 2016

- Implementation of CAN message sender/receiver on an Arduino device
- Construction of target hardware

Reference

EDUCATION

Technical University of Munich
October 2018 — August 2023

Bachelor of Science in Computer Science
Seminar work: The Evolution of the C++ Memory Model

Thesis: An Environment for Continuous Integration and Software Testing for sys-sage

SKILLS

C++20, Rust
JavaScript, Svelte, React
Java, Kotlin
Git, GitHub
WebGPU, Vulkan, Metal, OpenGL

Familiar: Swift, Android, Haskell

Languages: German, Czech, English C1, Spanish

Personal Interests: Realtime rendering, mathematics, Geometric Algebra, programming close to hardware, being productive

OPEN SOURCE CONTRIBUTIONS

egui (Rust Immediate-Mode UI)
Implement viewport dragging on macOS

iced (Rust UI library)
Implicit vertex generation for UI quads in a WGSL shader

winit (Rust windowing library)
Implement double tap gestures on macOS

stavec (Rust vector library)
Implement in-place swap-remove algorithm to achieve *O(1)* for element removal